Bethe Ansatz from Intersection Theory Viewpoint

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1 Introduction

This work is inspired by the report given by Yunfeng Jiang (https://www.koushare.com/live/details/11521).

2 Background

For a general introduction to the Bethe ansatz, we refer readers to [2]. The completeness problem of Bethe ansatz, which is our main concern, is discussed in [1].

In short, we aim to solve the following system of equations, known as the **Bethe equations**:

$$\left(\frac{\lambda_k + i/2}{\lambda_k - i/2}\right)^L = \prod_{j \neq k} \frac{\lambda_k - \lambda_j + i}{\lambda_k - \lambda_j - i}, \quad k = 1, \ldots, N$$

together with the auxiliary constraints listed below:

- The system is symmetric under permutations of λ_i; all roots obtained by permutation are regarded as equivalent root configurations.
- All λ_i are mutually distinct.
- If a root contains $\lambda_1 = i/2$, then necessarily $\lambda_2 = -i/2$. We further impose

$$\prod_{k=3,\ldots,N} \left(-\frac{\lambda_k + i/2}{\lambda_k - i/2}\right)^L = 1.$$

We assume $N < L/2$. If the Bethe ansatz is complete, the number of physical solutions equals

$$\binom{L}{N} - \binom{L}{N-1}.$$

Our core idea is as follows: for algebraic systems, once a rigorous definition of root multiplicity is established, the total number of counting solutions can be determined by the leading terms of the system via the **BKK theorem**. Moreover, such multiplicity coincides with the dimension of the corresponding Bethe vector.

We illustrate this viewpoint with a standard example:

$$\left(\frac{d^2}{dx^2} - t\right) f = 0,$$

whose characteristic equation reads $\lambda^2 - t = 0$. For $t \neq 0$, we have two distinct simple roots $\lambda = \pm\sqrt{t}$, each spanning a one-dimensional solution space $exp(\pm\sqrt{t}x)$. When $t = 0$, the root $\lambda = 0$ attains multiplicity 2, corresponding to a two-dimensional solution space that spanned by $1, x$.

Based on this analogy, we propose a example to illustrate the philosophy: suppose $(\lambda_1, \ldots, \lambda_N)$ solves the Bethe equations with $\lambda_1 = \lambda_2$ and all others distinct. If this degenerate root has algebraic multiplicity k, its real physical multiplicity should be $(k-1)/2$, that 1 is the non-physical multiplicity and 2 is the symmetric factor. One can construct $(k-1)/2$ valid Bethe vector starting from this configuration.

2.1 Roots at Infinity

Given any system of algebraic equations, the BKK theorem yields the **expected total number of complex solutions**, which equals the mixed volume of associated Newton polytopes, provided the coefficients satisfy generic open conditions.

We relax genericity conditions by requiring transversal intersection of zero loci of all defining polynomials. Compactification of $\mathbb{C}^n$ enables intersection theory on projective varieties: the Zariski closures of zero sets represent well-defined homology classes, and the expected solution count is given by the cup product of these homology classes. Solutions lying on the added divisors after compactification are termed **roots at infinity**.

3 Case $N = 2$

Define

$$f(\lambda_1, \lambda_2) = (\lambda_1 + i/2)^L(\lambda_1 - \lambda_2 - i) - (\lambda_1 - i/2)^L(\lambda_1 - \lambda_2 + i).$$

Set $x = \lambda_1, y = \lambda_2$. The Bethe equations reduce to

$$f(x, y) = 0, \qquad f(y, x) = 0.$$

We apply the BKK theorem by analyzing Newton polytopes via polynomial expansion:

- If L is even, the Newton polytope of $f(x, y)$ is the convex hull of vertices $(0, 0), (0, L), (1, L-1), (1, 1)$.

- If L is odd, the Newton polytope is the convex hull of $(0,0),(0,L),(1,L-1),(0,1)$.

The Newton polytope of $f(y,x)$ is the mirror image across the diagonal $x=y$. Direct computation shows the mixed volume equals L^2, so the full system admits L^2 complex solutions in total.

Next we count colliding solutions satisfying $x=y$. Substituting $x=y$ into the system gives a single polynomial equation of degree L, which has exactly L solutions. Consequently, the number of solutions with distinct roots is

$$\frac{L^2-L}{2}.$$

We then discuss the special root pair $(x=i/2, y=-i/2)$. We claim this root carries intrinsic multiplicity L arising from the structural derivation of Bethe equations. The actual dimension of physical Bethe vectors is equal to this algebraic multiplicity minus L, which is consistent with our prior conjectural counting rule.

3.1 Multiplicity of Root $(i/2,-i/2)$

Algebraic multiplicity is defined as

$$\dim_{\mathbb{C}}\left(\mathcal{O}_p/\mathcal{I}\right),$$

where notations follow standard definitions in GTM 52. Perform coordinate shifts:

$$x\mapsto x-i/2,\quad y\mapsto y+i/2.$$

The transformed system becomes

$$\begin{cases} f_1(x,y)=(x+i)^L(x-y)-x^L(x-y+2i),\\ f_2(x,y)=y^L(y-x-2i)-(y-i)^L(y-x).\end{cases}$$

The desired multiplicity is precisely the dimension of the local ring quotient $\mathbb{C}[[x,y]]/\langle f_1,f_2\rangle$.

Notice that f_1 is linear in y. Let $y_0\in\mathbb{C}[[x]]$ be the formal power series solution to $f_1(x,y_0)=0$, with expansion

$$y_0=x-2i^{-L+1}x^L+o(x^L).$$

Introduce new variable $y_{\text{new}}=y-y_0$, then factorize

$$f_1=\left(x^L-(x+i)^L\right)y_{\text{new}}.$$

The factor $x^L-(x+i)^L$ is invertible in the formal power series ring, so the ideal generated by f_1 coincides with $\langle y_{\text{new}}\rangle$.

We thus reduce the ring quotient to

$$\mathbb{C}[[x,y]]/\langle f_1,f_2\rangle\cong\mathbb{C}[[x]]/\langle f_2(x,y_0)\rangle.$$

Modulo x^{L+1}, we derive

$$f_2(x, y_0) \equiv -2i\big(1-(-1)^L\big)x^L.$$

For odd L, the multiplicity equals L; for even L, further expansion modulo x^{L+2} (to be determined) gives multiplicity $L+1$. This matches our prediction: the dimension of Bethe vectors associated with $(i/2, -i/2)$ differs from its algebraic multiplicity by exactly L.

4 Case $N = 3$

In this section we elaborate major difficulties encountered for $N = 3$. We adopt a modified inclusion-exclusion principle stated below.

Theorem 1. *Let $S \subset \mathbb{R}^3$ be a finite set with coordinates (x, y, z). Denote*

$$S_{x=y} = \{(x, y, z) \in S \mid x = y\}.$$

The number of elements in S with pairwise distinct coordinates is

$$\#S - \#S_{x=y} - \#S_{y=z} - \#S_{x=z} + 2\#S_{x=y=z}.$$

For general $N > 3$, see https://math.stackexchange.com/questions/3659385/.

Define

$$f(x, y, z) = \frac{1}{i}\Big[(x+i/2)^L(x-y-i)(x-z-i) - (x-i/2)^L(x-y+i)(x-z+i)\Big].$$

The polynomial f is symmetric in y, z, and the full Bethe system reads

$$f(x, y, z) = f(y, z, x) = f(z, x, y) = 0.$$

The leading homogeneous part of f is

$$(L-4)x^{L+1} - (L-2)(y+z)x^L + Lyzx^{L-1}.$$

Vertices of its Newton polytope are listed as:

$$(1,0,0),\ (0,1,0),\ (0,0,1),\ (1,1,1),\ (1,1,L-1),\ (1,0,L),\ (0,1,L),\ (0,0,L+1).$$

By BKK counting:

- The full system has $(L+1)^3$ complex solutions;
- The constrained system $f(x, y, z) = f(z, x, y) = x - y = 0$ has $(L+1)^2$ solutions;
- The fully colliding system $f(x, y, z) = x-z = x-y = 0$ has $L+1$ solutions.

Substitute into inclusion-exclusion formula, and we obtain $\binom{L+1}{3}$ solutions with pairwise distinct roots.

4.1 Multiplicity of Roots of Type $(i/2, -i/2, z)$

Substitute $x = i/2, y = -i/2$ into the system, then $f(x, y, z) = f(y, z, x) = 0$ holds identically. We factorize

$$f(z, x, y) = (z - i/2)(z + i/2) \cdot \left(z^{L-1} + \cdots\right),$$

which yields $L - 1$ nontrivial roots excluding $z = \pm i/2$. We conjecture each such root carries excess multiplicity L. After subtracting all non-physical contributions, the valid physical solution number becomes

$$\binom{L+1}{3} - L(L-1) = \binom{L}{3} - \binom{L}{2},$$

which coincides with the completeness formula.

Remark 1. *Several implicit assumptions are imposed to validate the above arguments:*

- *Generic open conditions required by the BKK theorem fail for actual Bethe equations. For instance, set $L = 6, N = 3$ and impose triple collision $x = y = z$, we obtain a reduced polynomial $f = 3x^5 - \frac{5}{2}x^3 + \frac{3}{16}x$, while naive BKK counting predicts 7 solutions. Nevertheless, our combinatorial counting still holds, since all roots at infinity lie within collision hyperplanes $\lambda_i = \lambda_j$ and make no contribution to physical spectra.*
- *The algebraic multiplicity of the degenerate root $(-i/2, i/2, i/2)$ remains unchanged when restricted to the subvariety defined by $z - y = 0$.*

5 General Case $N \geq 4$

For $N = 4$, BKK theorem gives total formal solution count $\binom{L+2}{4}$. Supported by numerical evidences, we propose the following conjectures:

- Configurations of the form $(-i/2, i/2, z, w)$ with $z, w \notin \{\pm i/2\}$ contribute $\binom{L-1}{2}$ non-physical solutions;
- Configurations $(-i/2, i/2, i/2, w)$ with $w \notin \{\pm i/2\}$ carry no extra effective multiplicity;
- The fully paired degenerate root $(-i/2, -i/2, i/2, i/2)$ has total multiplicity $\binom{L}{2}$.

After removing all non-physical contributions, the number of physical solutions is

$$\binom{L+2}{4} - L \cdot \binom{L-1}{2} - \binom{L}{2} = \binom{L}{4} - \binom{L}{3},$$

consistent with the completeness formula.

Define the standard completeness number

$$S(L,N) = \binom{L}{N} - \binom{L}{N-1}.$$

For general positive integers L, N with $L \geq 2N$, we expect all formal complex solutions counted by BKK can be decomposed into physical solutions plus degenerate non-physical ones with extra multiplicity.

Let $B(k;L,N)$ denote the number of degenerate root configurations and $M(k;L,N)$ denote the corresponding excess multiplicity. We postulate the decomposition formula:

$$S(L,N) = \binom{L+N-2}{N} - \sum_{k=1}^{\lfloor L/2 \rfloor} B(k;L,N) \cdot M(k;L,N).$$

Early ansätze $B(k;L,N) = \binom{L-1}{N-2k}$ and $M(k;L,N) = \binom{L}{k}$ turn out invalid already for $N = 6$.

6 Overall Proof Strategy

Let L, N be positive integers satisfying $L \geq 2N$. Define the core polynomial

$$f(\lambda_1, \dots, \lambda_N) = (\lambda_1 + i/2)^L \prod_{j \neq 1} (\lambda_1 - \lambda_j + i) - (\lambda_1 - i/2)^L \prod_{j \neq 1} (\lambda_1 - \lambda_j - i).$$

f is symmetric in the last $N-1$ variables. The full Bethe system is written as cyclic permutations:

$$\begin{cases} f(\lambda_1, \lambda_2, \dots, \lambda_N) = 0, \\ f(\lambda_2, \lambda_1, \dots, \lambda_N) = 0, \\ \quad \vdots \\ f(\lambda_N, \lambda_1, \dots, \lambda_{N-1}) = 0. \end{cases}$$

For every complex root configuration $\boldsymbol{\lambda} = (\lambda_1, \dots, \lambda_N)$, we introduce the concept of **non-physical multiplicity**. The dimension of genuine physical Bethe vectors equals the raw algebraic multiplicity within Bethe equations minus this non-physical multiplicity.

References

[1] W. Hao, R. I. Nepomechie and A. J. Sommese, Completeness of solutions of Bethe's equations, Phys. Rev. E **88** (2013), 052113, arXiv:1308.4645.

[2] N. A. Slavnov, Algebraic Bethe ansatz, 2018.